Methods to Detect AI-Generated Videos

1. Metadata Inspection (Fast)

Check:

- Camera Make/Model
- GPS
- Frame rate
- Codec
- Editing software

If metadata says

Runway
Synthesia
Pika
Kling
Veo

then the video was probably AI-generated.

Libraries:

- FFmpeg
- ExifTool
- MediaInfo

2. Deepfake Detection Models

Use models trained specifically for AI-generated video detection.

Examples include:

- Microsoft's Video Authenticator
- DeepFakeBench
- FaceForensics++
- OpenForensics
- X-CLIP-based deepfake detectors

These models analyze:

- Face inconsistencies
- Lighting
- Eye blinking
- Skin texture
- Compression artifacts
- Temporal consistency between frames

Output:

Real: 97%

AI Generated: 3%

3. Frame-Level Detection

Extract frames every second.

Video

↓
Frame 1
Frame 2
Frame 3
...

Run an AI-image detector on each frame.

If most frames appear AI-generated:

Result:
AI probability = 91%

4. Motion Consistency Analysis

AI videos often have:

- unnatural body movements
- inconsistent limb lengths
- flickering
- changing backgrounds
- impossible physics

Pose estimation models like OpenPose or MediaPipe can detect abnormal motion patterns.

5. Watermark Detection

Some AI video generators embed invisible or visible watermarks.

Check for:

- Google Veo
- Synthesia
- Runway
- Pika
- Kling

Invisible watermark detection is supported by some providers.

Athlete Upload
↓
Metadata Check
↓
Deepfake Detector
↓
Frame AI Detector
↓
Motion Consistency Check
↓
Human Review (if confidence is uncertain)
↓
GPT Performance Analysis